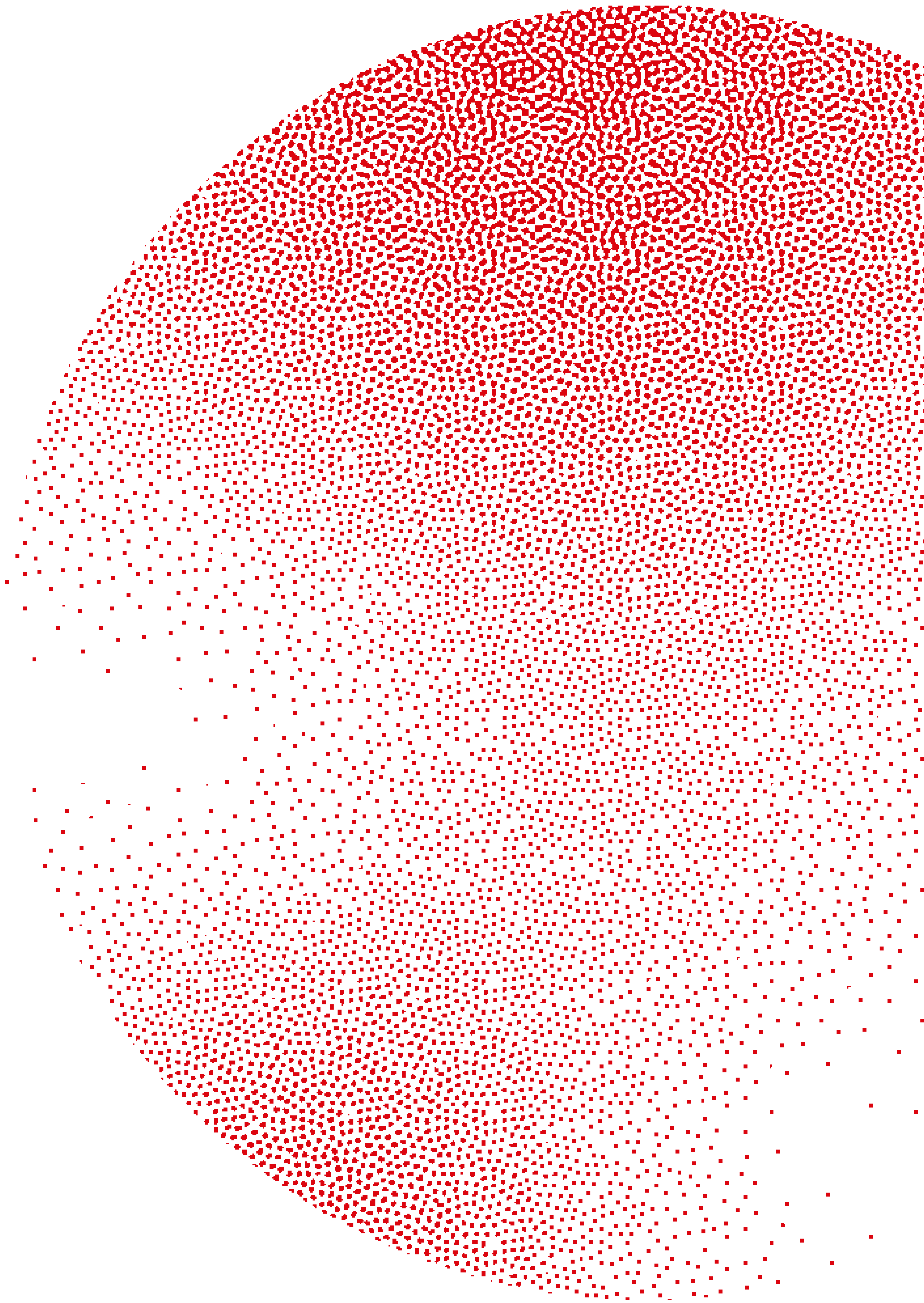
Swiss Institute of Bioinformatics

Metagenomic Data Analysis of Microbial Communities (MGMC01)

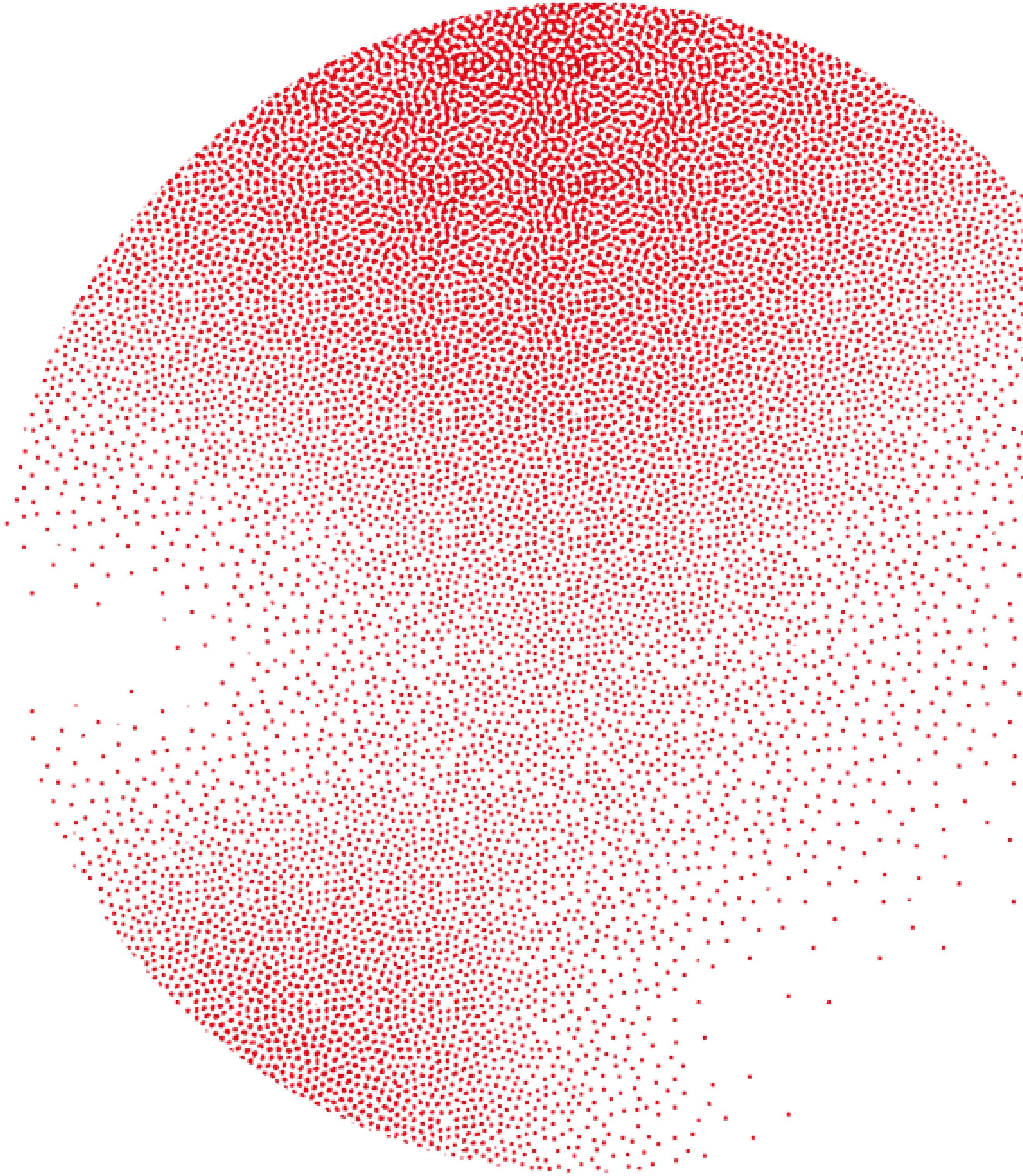
Day 3 - Session 1

PhD Jeferyd Yepes-García
University of Fribourg
October 28th, 2026



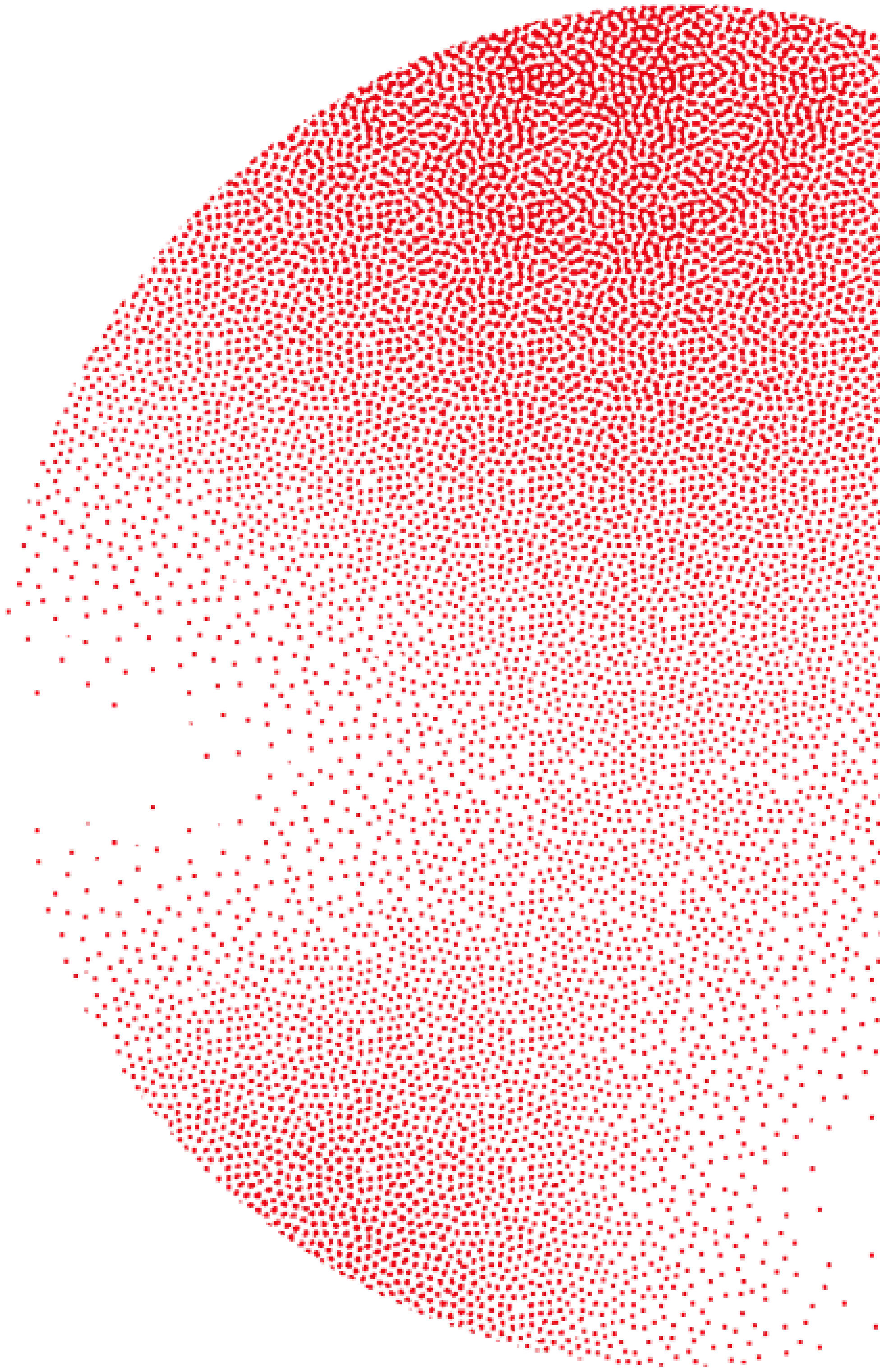
This presentation

- Learning outcomes
- Why metagenomics?
- Contaminant removal
- Taxonomic classification/profiling
- Usage of long reads
- Summary

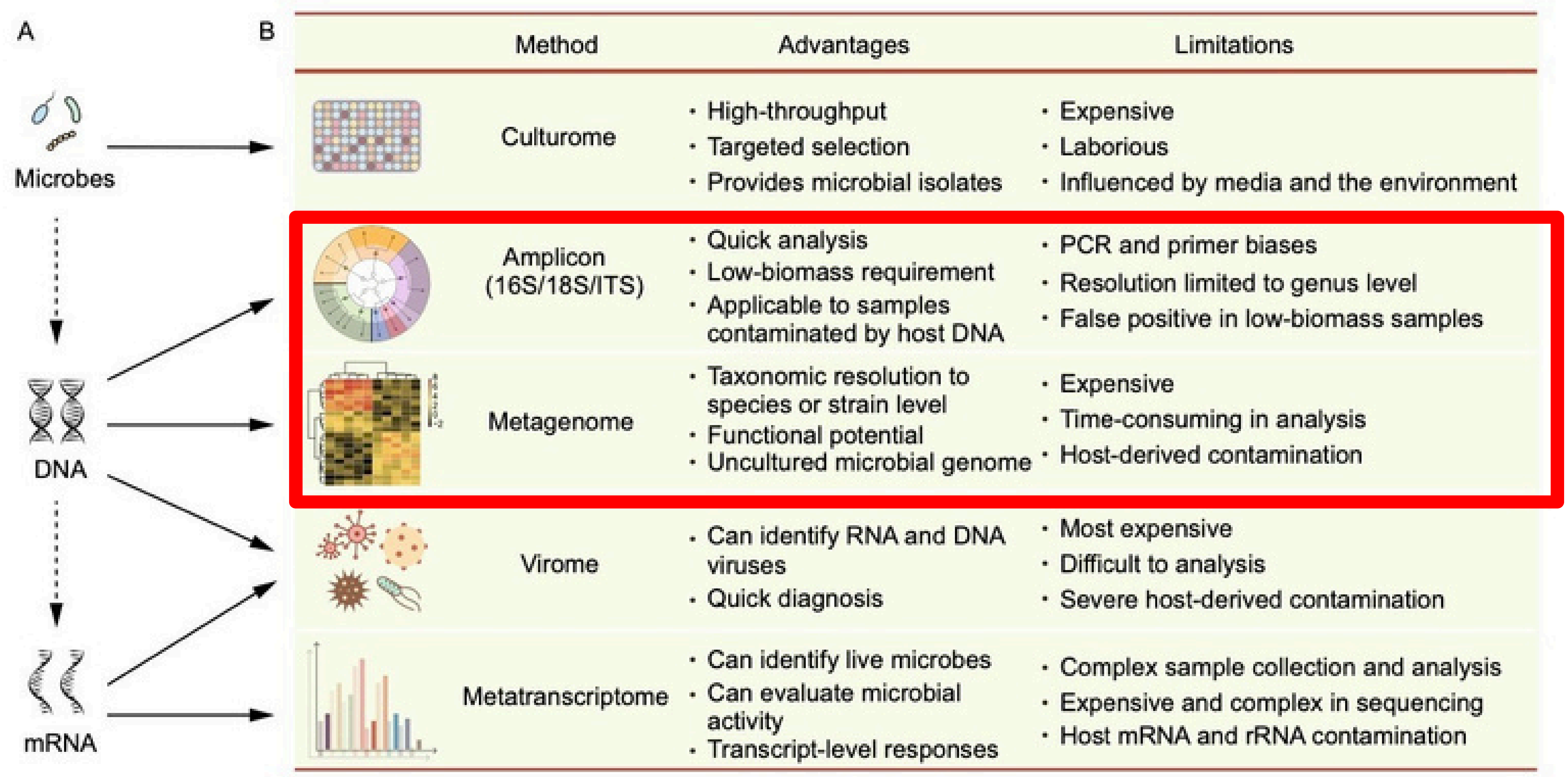


Learning outcomes

- Determine why performing metagenomics is a good methodological choice under certain scenarios
- Recognize why it is necessary to remove contaminant/host reads to enhance the quality of the analysis.
- Understand the main principle behind taxonomic classification of unassembled reads.
- Execute a complete workflow to perform taxonomic classification and carry out a preliminary analsysis.

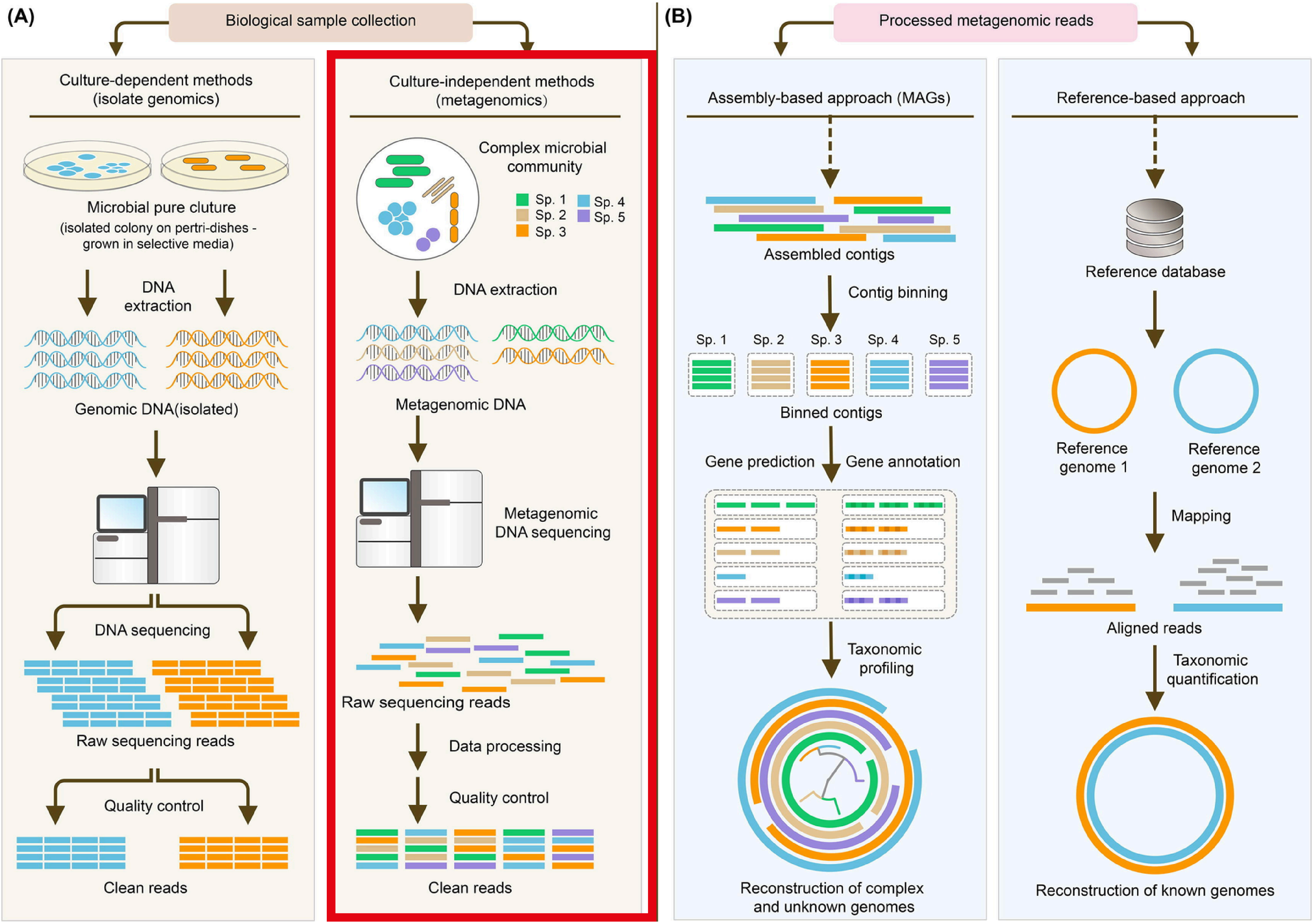


How to study a microbiome/metagenome?


A
B
Microbes
DNA
mRNA

Method	Advantages	Limitations
Culturome	- High-throughput - Targeted selection - Provides microbial isolates	- Expensive - Laborious - Influenced by media and the environment
Amplicon (16S/18S/ITS)	- Quick analysis - Low-biomass requirement - Applicable to samples contaminated by host DNA	- PCR and primer biases - Resolution limited to genus level - False positive in low-biomass samples
Metagenome	- Taxonomic resolution to species or strain level - Functional potential - Uncultured microbial genome	- Expensive - Time-consuming in analysis - Host-derived contamination
Virome	- Can identify RNA and DNA viruses - Quick diagnosis	- Most expensive - Difficult to analysis - Severe host-derived contamination
Metatranscriptome	- Can identify live microbes - Can evaluate microbial activity - Transcript-level responses	- Complex sample collection and analysis - Expensive and complex in sequencing - Host mRNA and rRNA contamination

How to study a microbiome/metagenome?


(A)
Biological sample collection
Culture-dependent methods
(isolate genomics)
Microbial pure cluture
(isolated colony on pertri-dishes - grown in selective media)
DNA extraction
Genomic DNA(isolated)
DNA sequencing
Raw sequencing reads
Quality control
Clean reads
Culture-independent methods
(metagenomics)
Complex microbial community
Sp. 1
Sp. 2
Sp. 3
Sp. 4
Sp. 5
DNA extraction
Metagenomic DNA
Metagenomic DNA sequencing
Raw sequencing reads
Data processing
Quality control
Clean reads
(B)
Processed metagenomic reads
Assembly-based approach (MAGs)
Assembled contigs
Contig binning
Sp. 1
Sp. 2
Sp. 3
Sp. 4
Sp. 5
Binned contigs
Gene prediction
Gene annotation
Taxonomic profiling
Reconstruction of complex and unknown genomes
Reference-based approach
Reference database
Reference genome 1
Reference genome 2
Mapping
Aligned reads
Taxonomic quantification
Reconstruction of known genomes

Why metagenomics?

Shotgun sequencing of all DNA in a sample

01
Culture-independent
Access microbes that cannot be cultured

02
Functional potential
Genes and pathways, not only who is there

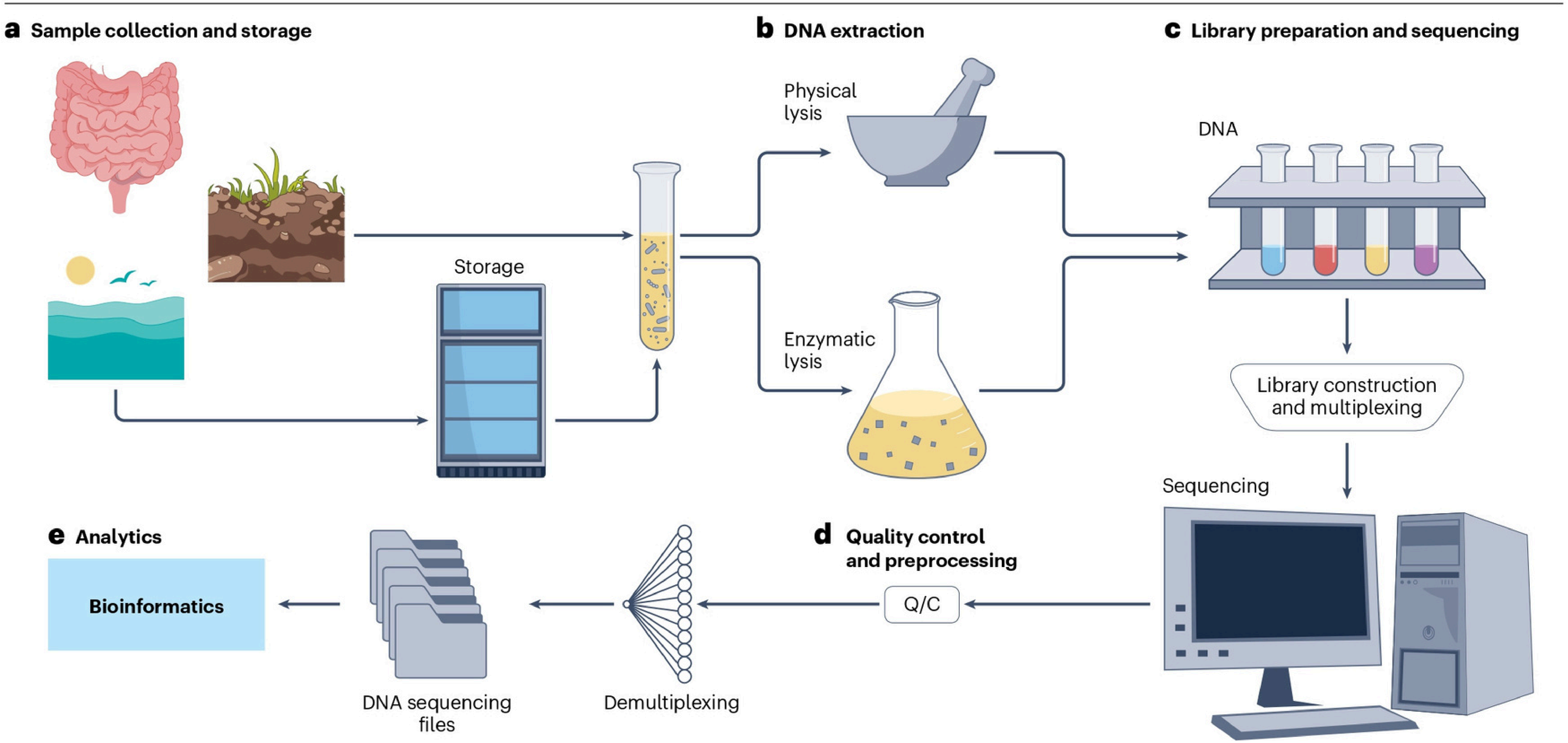
03
Higher resolution
Species- and strain-level profiling

04
All domains at once
Bacteria, archaea, viruses and fungi in one assay

05
No primer bias
No marker-gene PCR amplification step

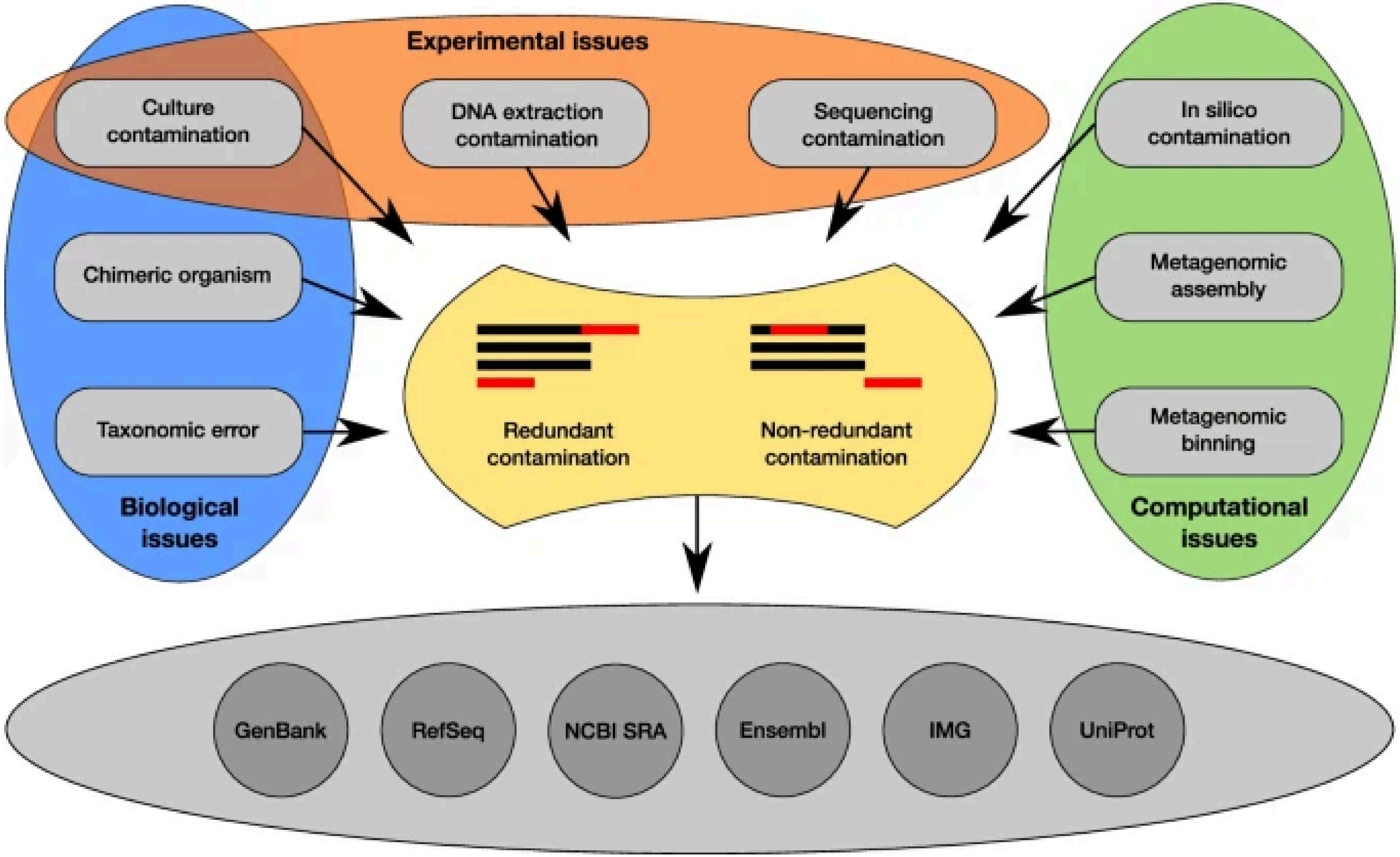
06
Genome recovery
Assemble MAGs and discover novel genes

Metagenomics project workflow:


a Sample collection and storage
Storage
b DNA extraction
Physical lysis
Enzymatic lysis
c Library preparation and sequencing
DNA
Library construction and multiplexing
Sequencing
d Quality control and preprocessing
Q/C
Demultiplexing
DNA sequencing files
e Analytics
Bioinformatics

Liu et al. (2025)

What are contaminants?

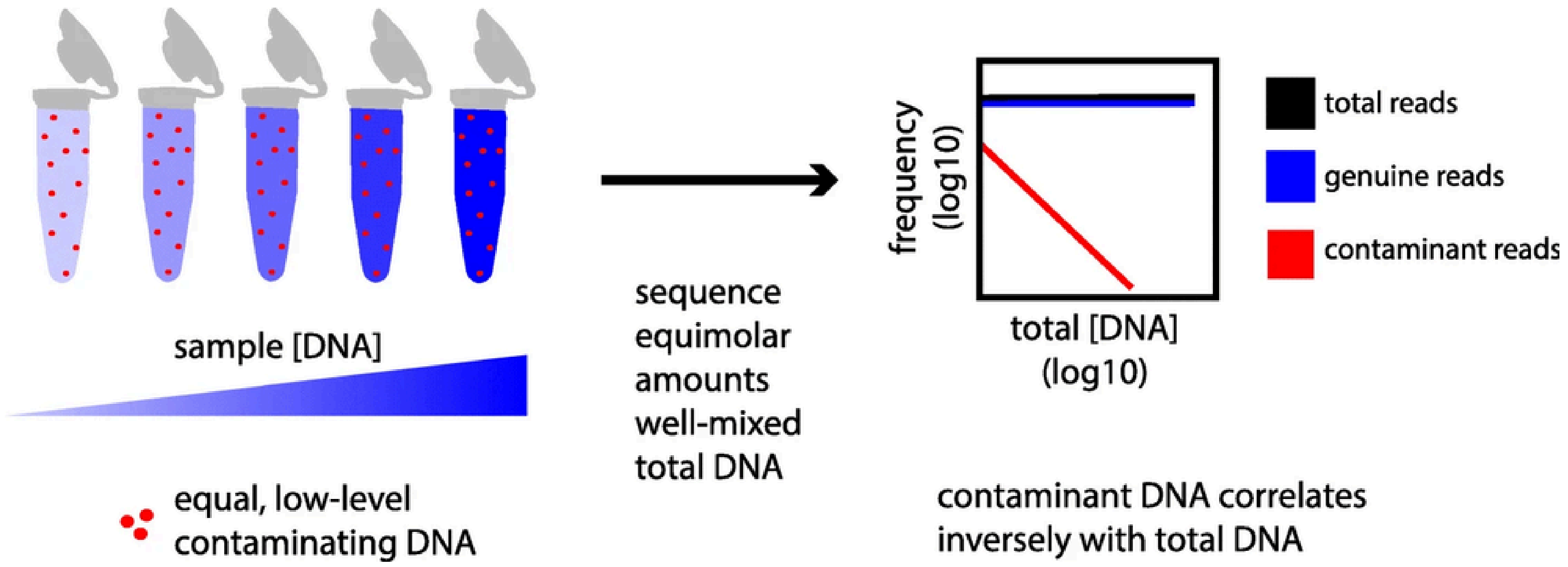

Experimental issues
Culture contamination
DNA extraction contamination
Sequencing contamination
In silico contamination
Chimeric organism
Metagenomic assembly
Taxonomic error
Metagenomic binning
Redundant contamination
Non-redundant contamination
Biological issues
Computational issues
GenBank
RefSeq
NCBI SRA
Ensembl
IMG
UniProt

Cornet & Baurain (2021)

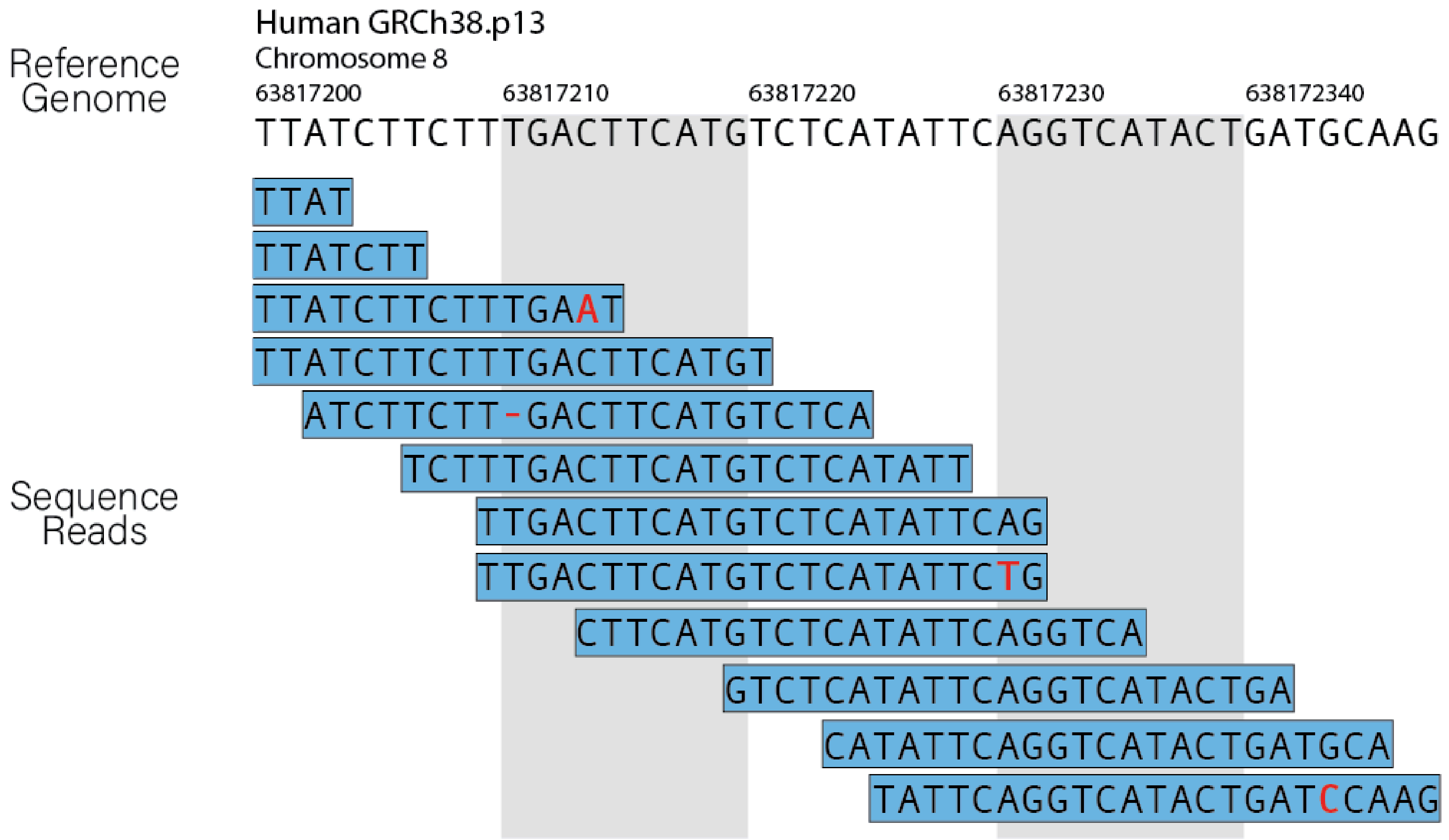
Problems caused by contaminants:

- **Misleading Taxonomic Profiles:** False positives, inflated diversity**.**
- **Erroneous Functional Profiles:** Misassignment/overestimation of functions**.**
- **Biased Abundance Estimates:** Skewed abundances, false dominance**.**
- **Increased Computational Load:** Resource waste, algorithm performance**.**
- **Money thrown away.**
- **Reproducibility issues.**

Tools to remove sequences:

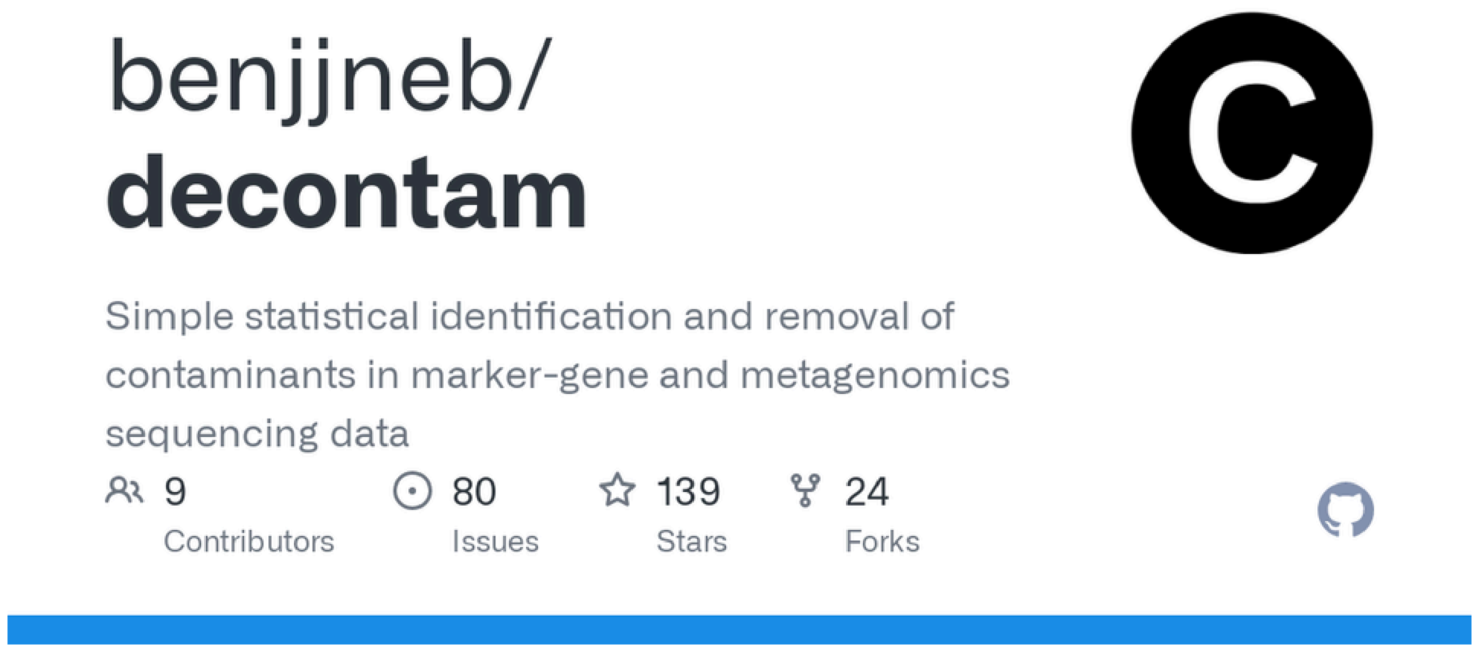

sample [DNA]
equal, low-level contaminating DNA
sequence equimolar amounts well-mixed total DNA
frequency (log10)
total [DNA] (log10)
total reads
genuine reads
contaminant reads
contaminant DNA correlates inversely with total DNA

Statistical approach


Reference Genome
Human GRCh38.p13
Chromosome 8
63817200
63817210
63817220
63817230
638172340
TTATCTTCTTTGACTTCATGTCTCATATTCAGGTCATACTGATGCAAG
Sequence Reads
TTAT
TTATCTT
TTATCTTCTTTGAAT
TTATCTTCTTTGACTTCATGT
ATCTTCTT-GACTTCATGTCTCA
TCTTTGACTTCATGTCTCATATT
TTGACTTCATGTCTCATATTCAG
TTGACTTCATGTCTCATATTCTG
CTTCATGTCTCATATTCAGGTCA
GTCTCATATTCAGGTCATACTGA
CATATTCAGGTCATACTGATGCA
TATTCAGGTCATACTGATCCAAG

Alignment against indexed genomes

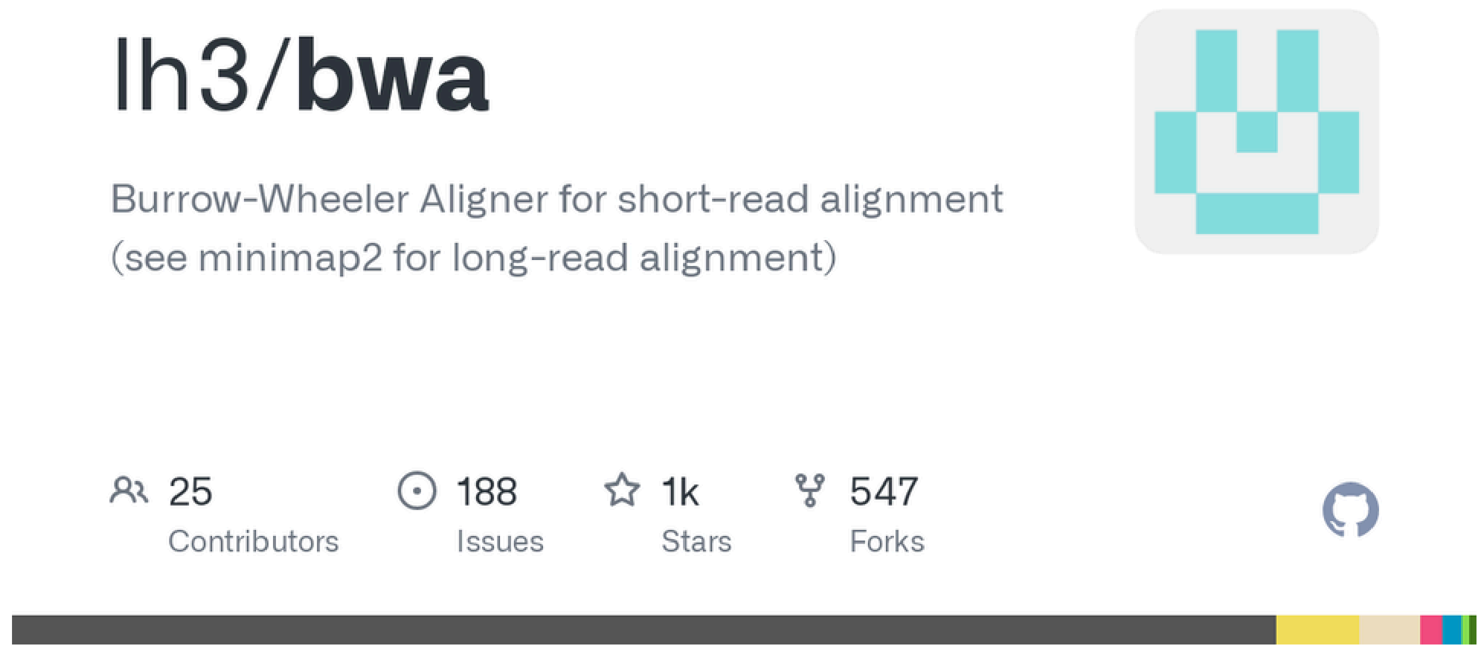
Tools to remove sequences:



benjjneb/
decontam

Simple statistical identification and removal of contaminants in marker-gene and metagenomics sequencing data

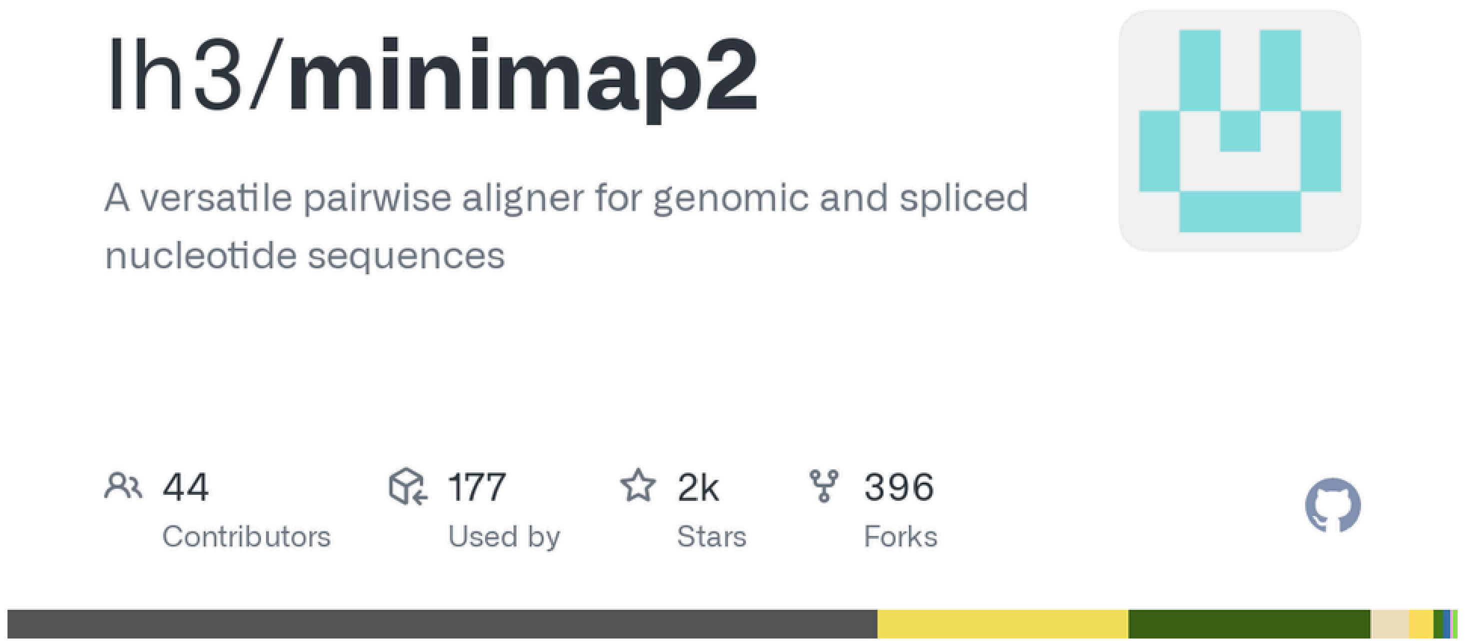
9 Contributors · 80 Issues · 139 Stars · 24 Forks



lh3/**bwa**

Burrow-Wheeler Aligner for short-read alignment (see minimap2 for long-read alignment)

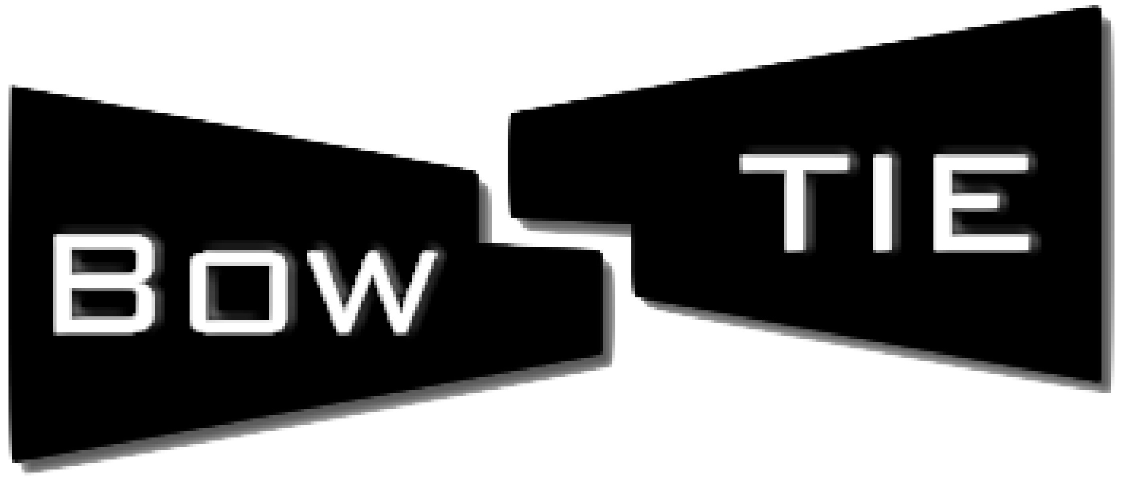
25 Contributors · 188 Issues · 1k Stars · 547 Forks



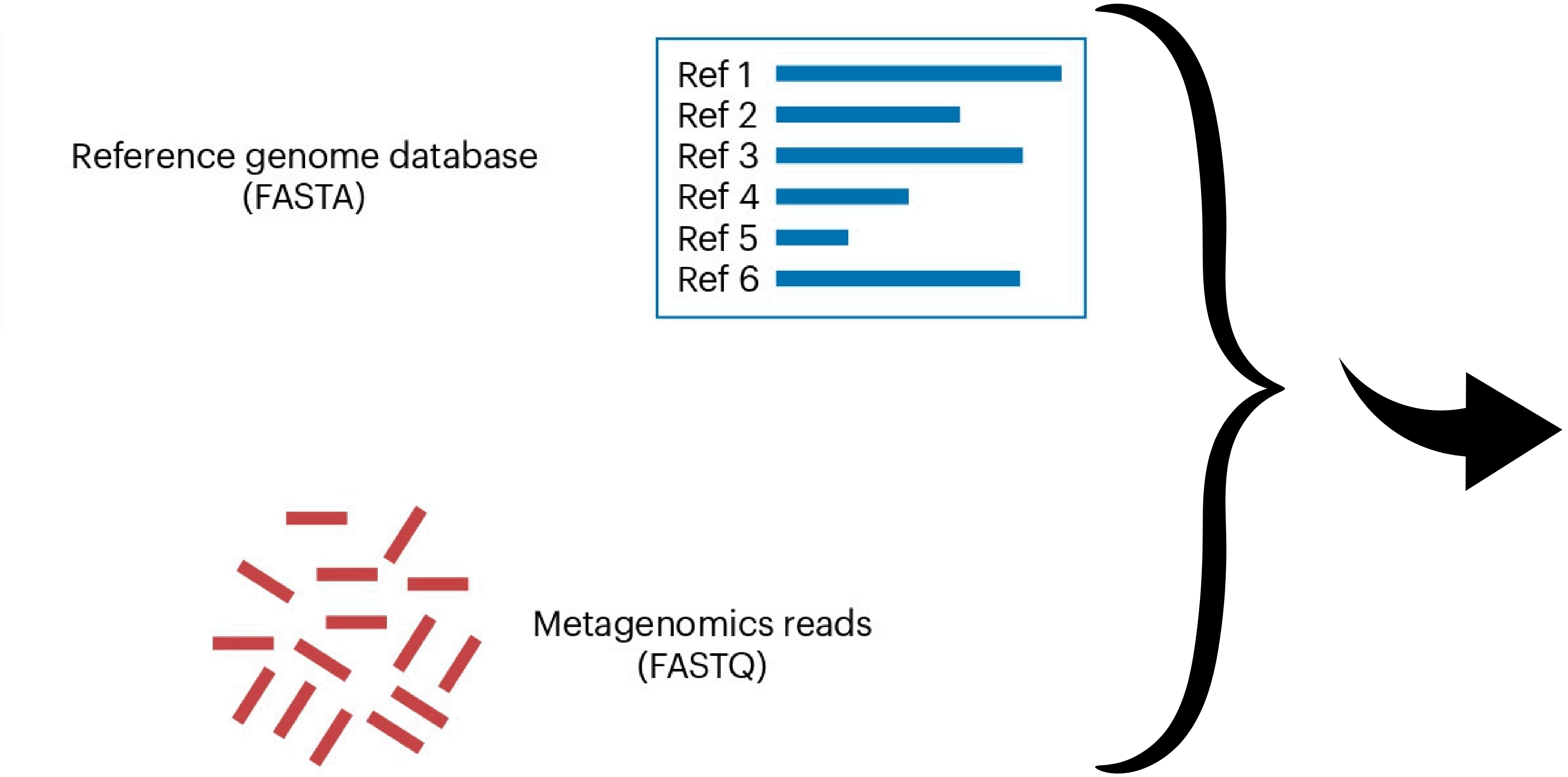
lh3/**minimap2**

A versatile pairwise aligner for genomic and spliced nucleotide sequences

44 Contributors · 177 Used by · 2k Stars · 396 Forks

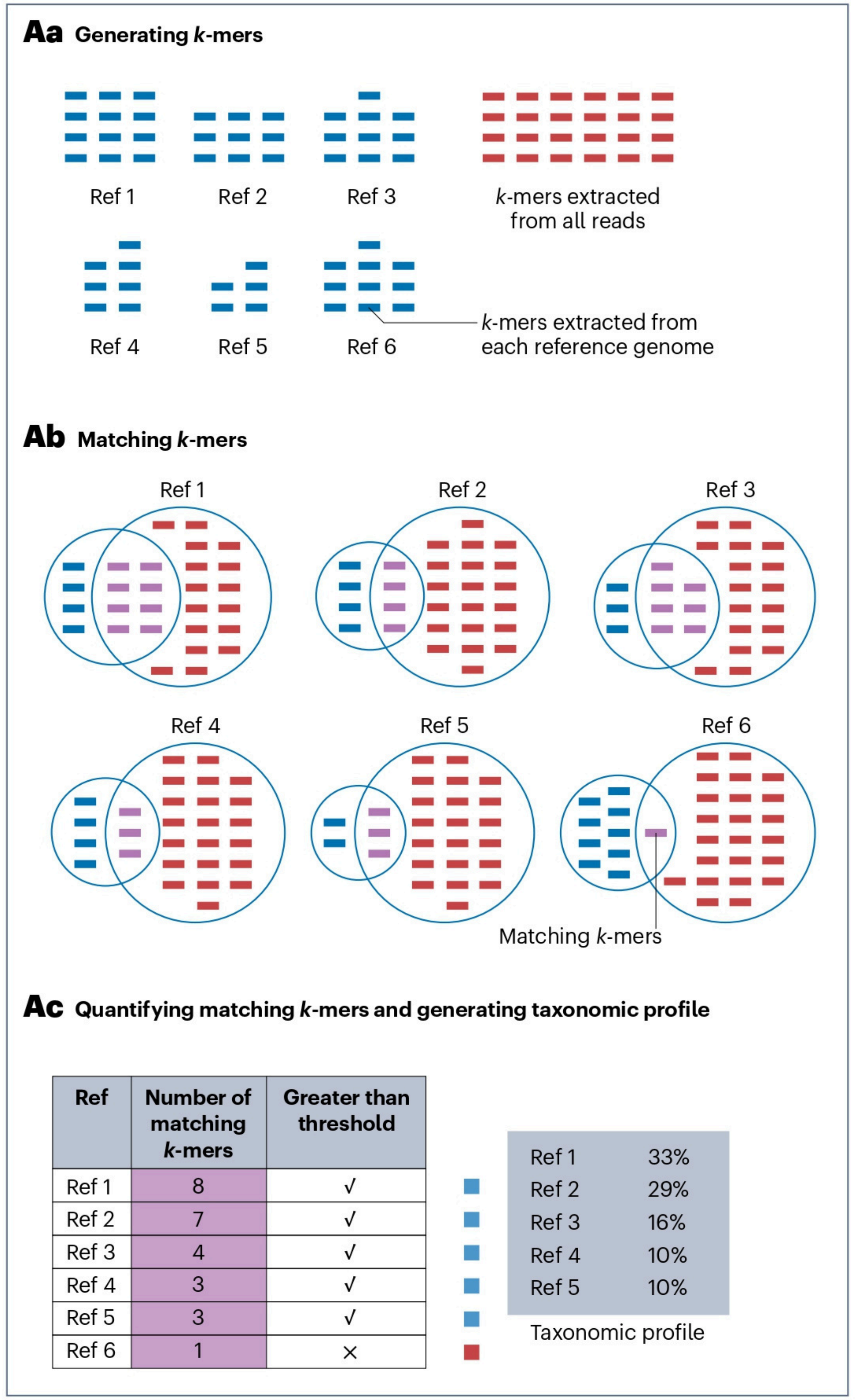


Classification vs profiling:


Ref 1
Ref 2
Ref 3
Ref 4
Ref 5
Ref 6

Reference genome database
(FASTA)

Metagenomics reads
(FASTQ)


Aa **Generating *k*-mers**
Ref 1
Ref 2
Ref 3
k-mers extracted from all reads
Ref 4
Ref 5
Ref 6
k-mers extracted from each reference genome
Ab **Matching *k*-mers**
Ref 1
Ref 2
Ref 3
Ref 4
Ref 5
Ref 6
Matching *k*-mers
Ac **Quantifying matching *k*-mers and generating taxonomic profile**
Taxonomic profile

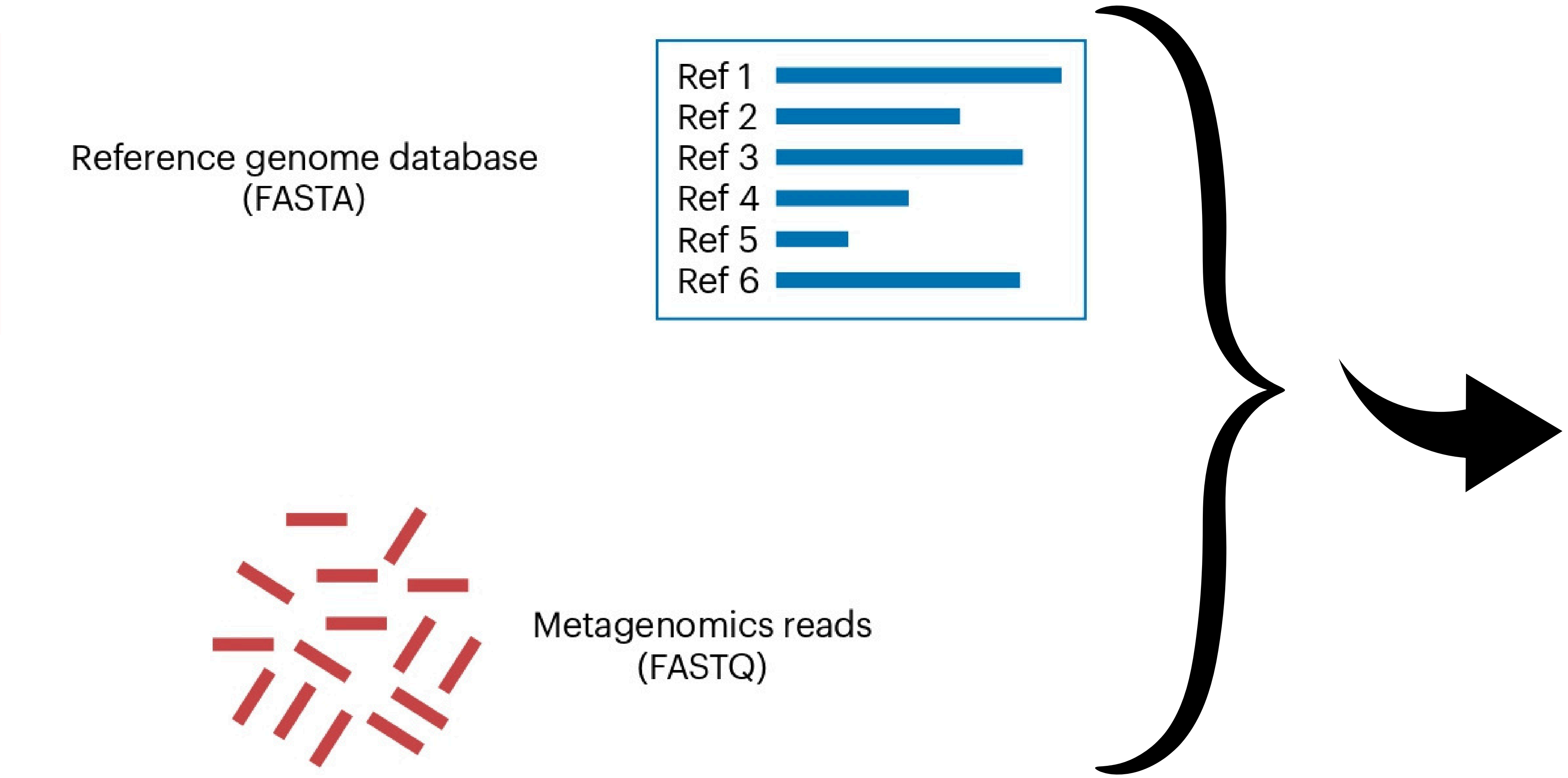
Ref	Number of matching *k*-mers	Greater than threshold
Ref 1	8	√
Ref 2	7	√
Ref 3	4	√
Ref 4	3	√
Ref 5	3	√
Ref 6	1	×

🟦	Ref 1	33%
🟦	Ref 2	29%
🟦	Ref 3	16%
🟦	Ref 4	10%
🟦	Ref 5	10%
🟥		

Taxonomic profile

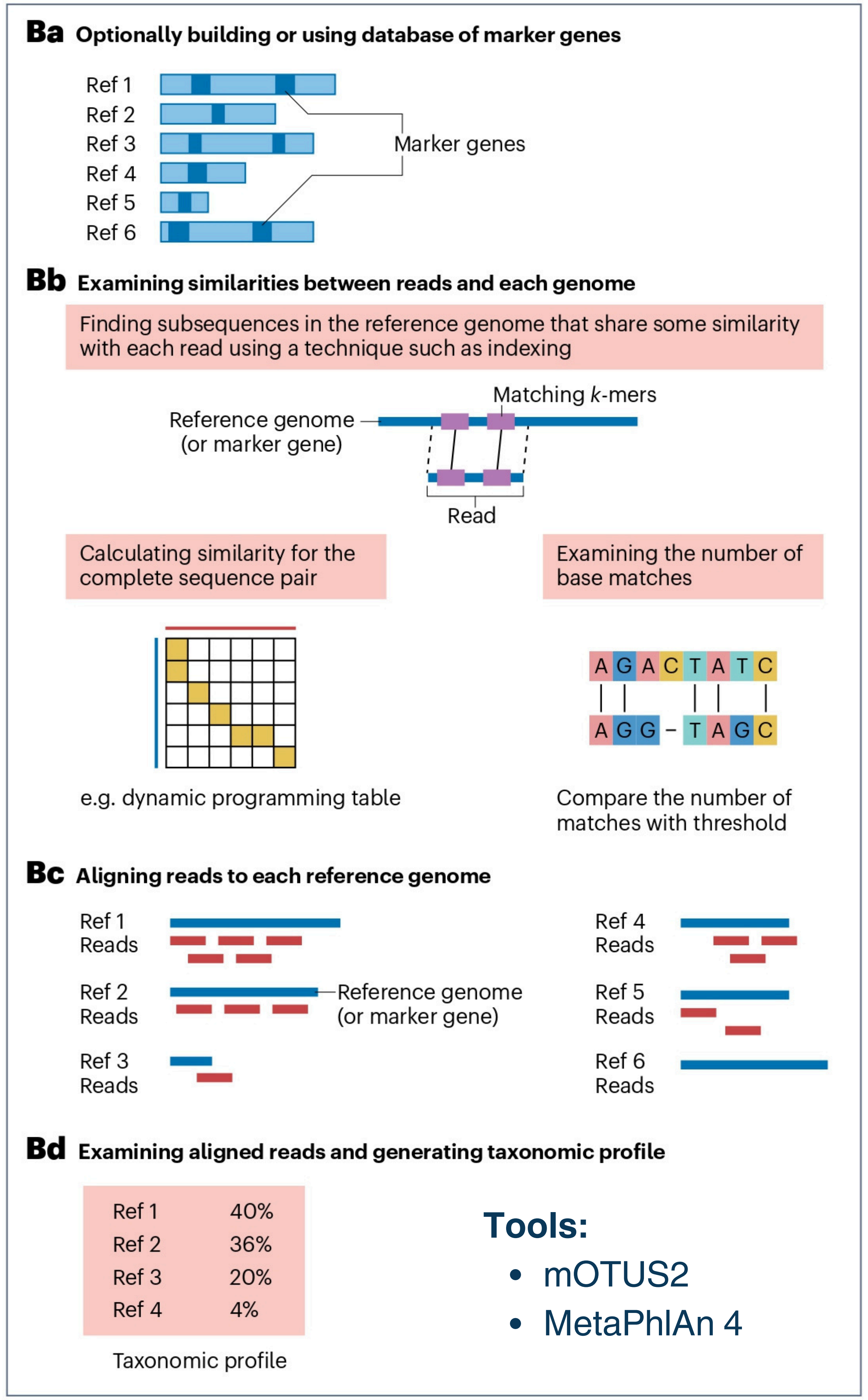
Liu et al. (2025)

Classification **vs profiling**:


Ref 1
Ref 2
Ref 3
Ref 4
Ref 5
Ref 6

Reference genome database
(FASTA)

Metagenomics reads
(FASTQ)

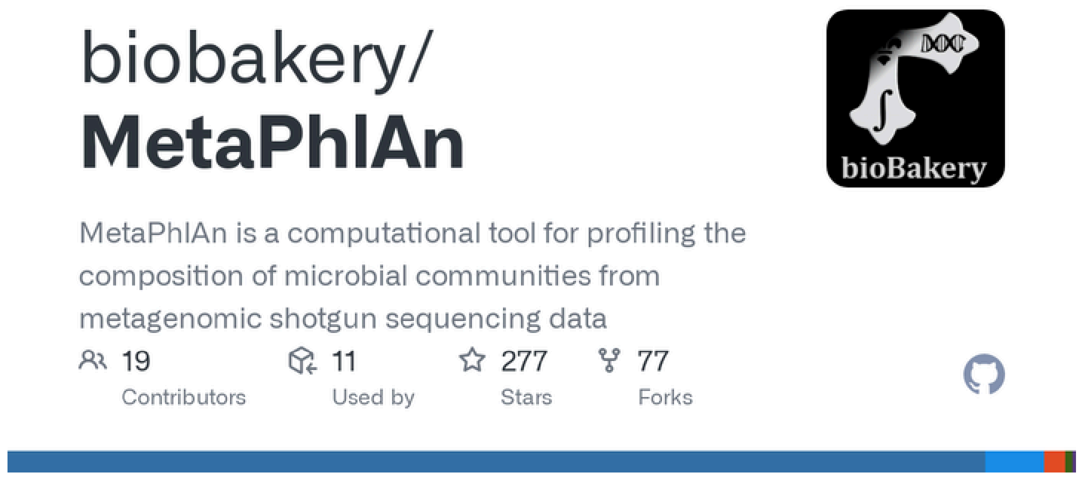

Ba Optionally building or using database of marker genes
Ref 1
Ref 2
Ref 3
Ref 4
Ref 5
Ref 6
Marker genes
Bb Examining similarities between reads and each genome
Finding subsequences in the reference genome that share some similarity with each read using a technique such as indexing
Matching k-mers
Reference genome (or marker gene)
Read
Calculating similarity for the complete sequence pair
Examining the number of base matches
A G A C T A T C
A G G – T A G C
e.g. dynamic programming table
Compare the number of matches with threshold
Bc Aligning reads to each reference genome
Ref 1 Reads
Ref 2 Reads
Ref 3 Reads
Ref 4 Reads
Ref 5 Reads
Ref 6 Reads
Reference genome (or marker gene)
Bd Examining aligned reads and generating taxonomic profile
Ref 1 40%
Ref 2 36%
Ref 3 20%
Ref 4 4%
Taxonomic profile

Tools:

- mOTUS2
- MetaPhlAn 4

Liu et al. (2025)

Taxonomic profiling/classification tools

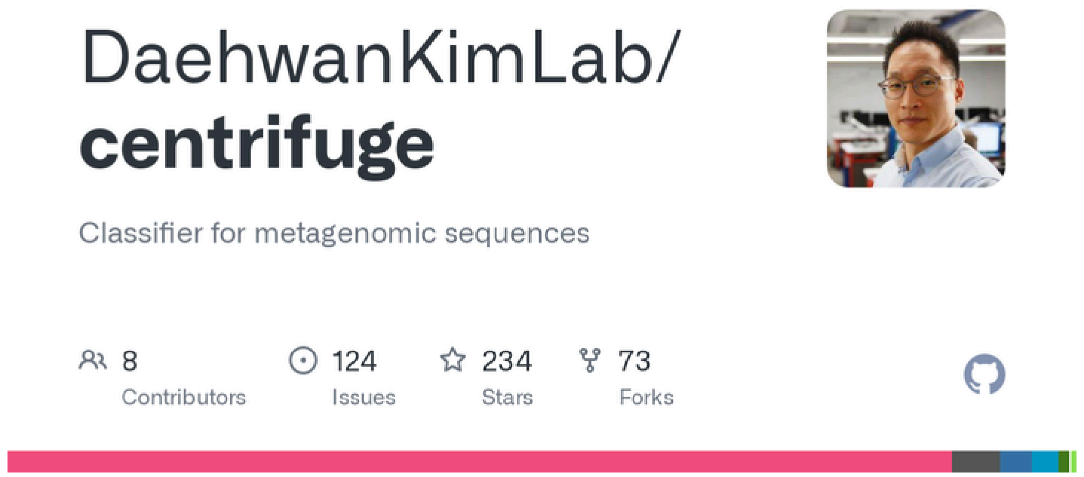


biobakery/
MetaPhlAn

MetaPhlAn is a computational tool for profiling the composition of microbial communities from metagenomic shotgun sequencing data

bioBakery
19 Contributors
11 Used by
277 Stars
77 Forks

Unique clade-specific marker genes

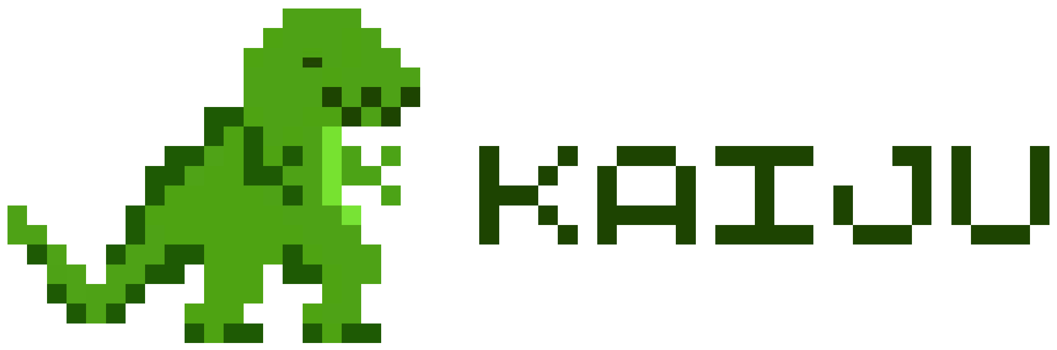


DaehwanKimLab/
centrifuge

Classifier for metagenomic sequences

8 Contributors
124 Issues
234 Stars
73 Forks

Indexed genomes (BWT and FM)



Reference protein database

MEGAN6
Analysis and visualization

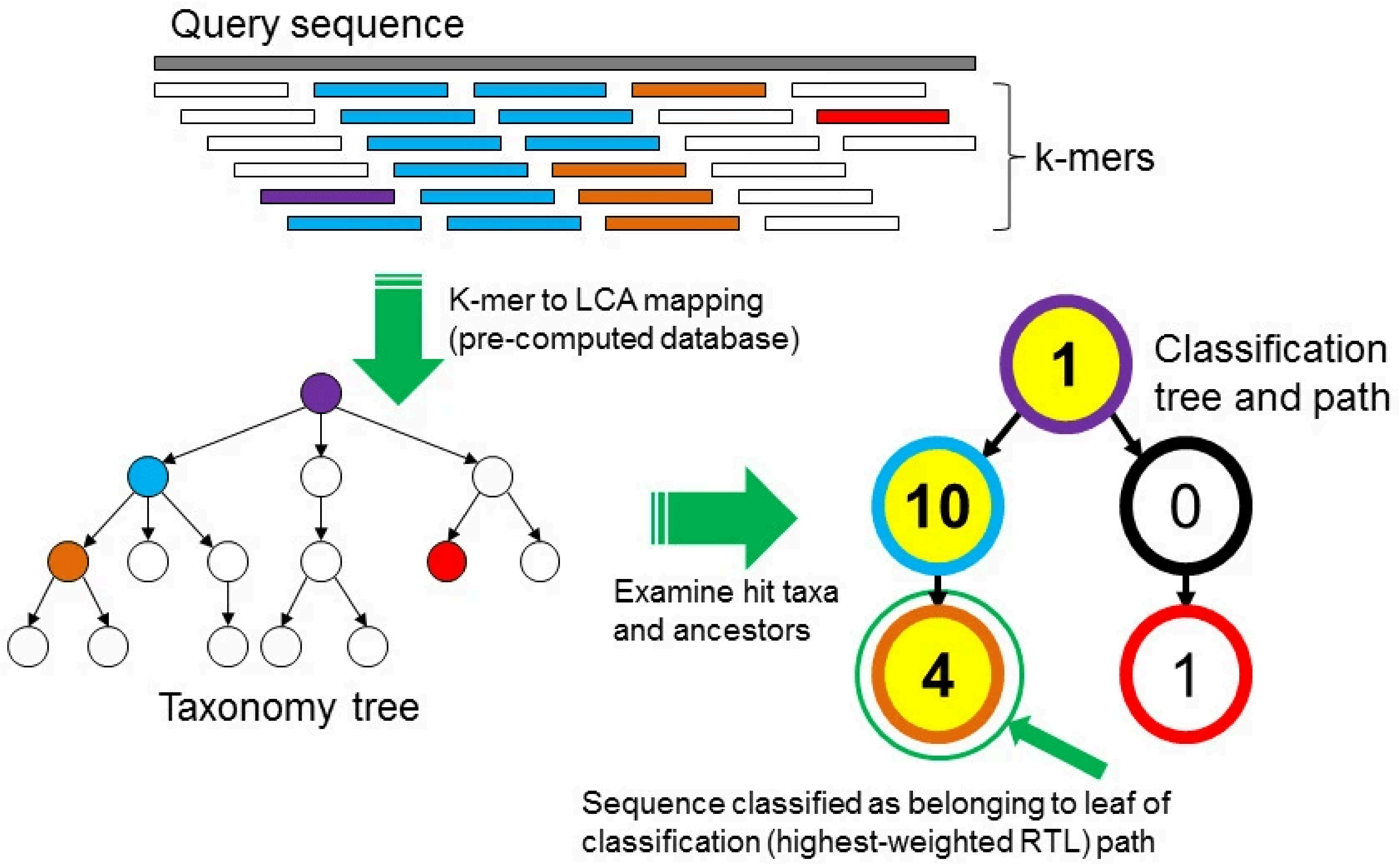
NCBI or SILVA taxonomy

CLARK

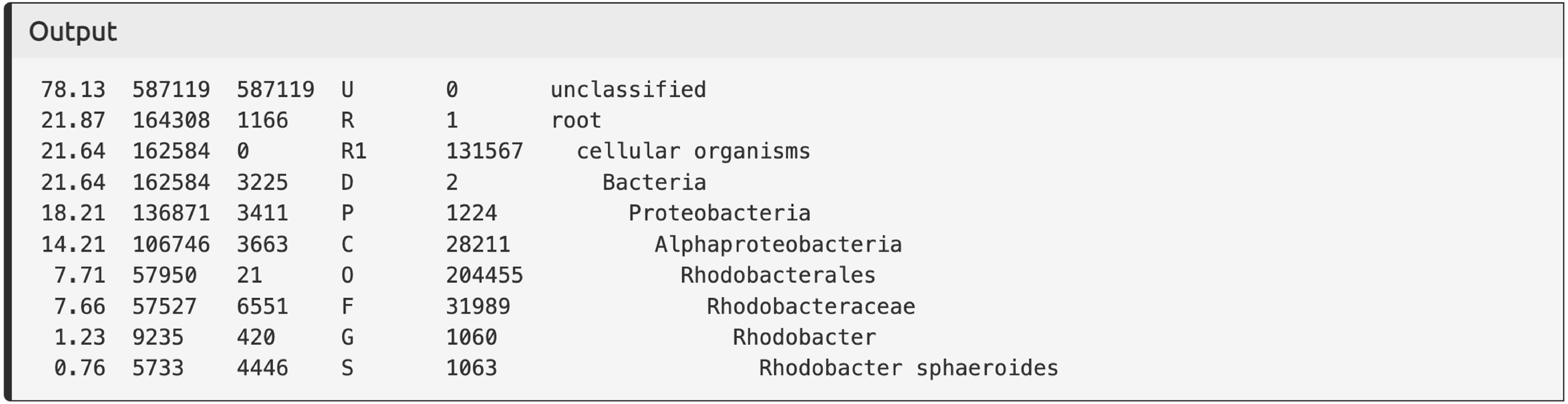
Fast, accurate and versatile sequence classification system

Discriminative k-mers

Kraken2: Exact k-mer matches.


Query sequence
k-mers
K-mer to LCA mapping
(pre-computed database)
Taxonomy tree
Examine hit taxa
and ancestors
Classification
tree and path
1
10
0
4
1
Sequence classified as belonging to leaf of
classification (highest-weighted RTL) path

Kraken2: Output

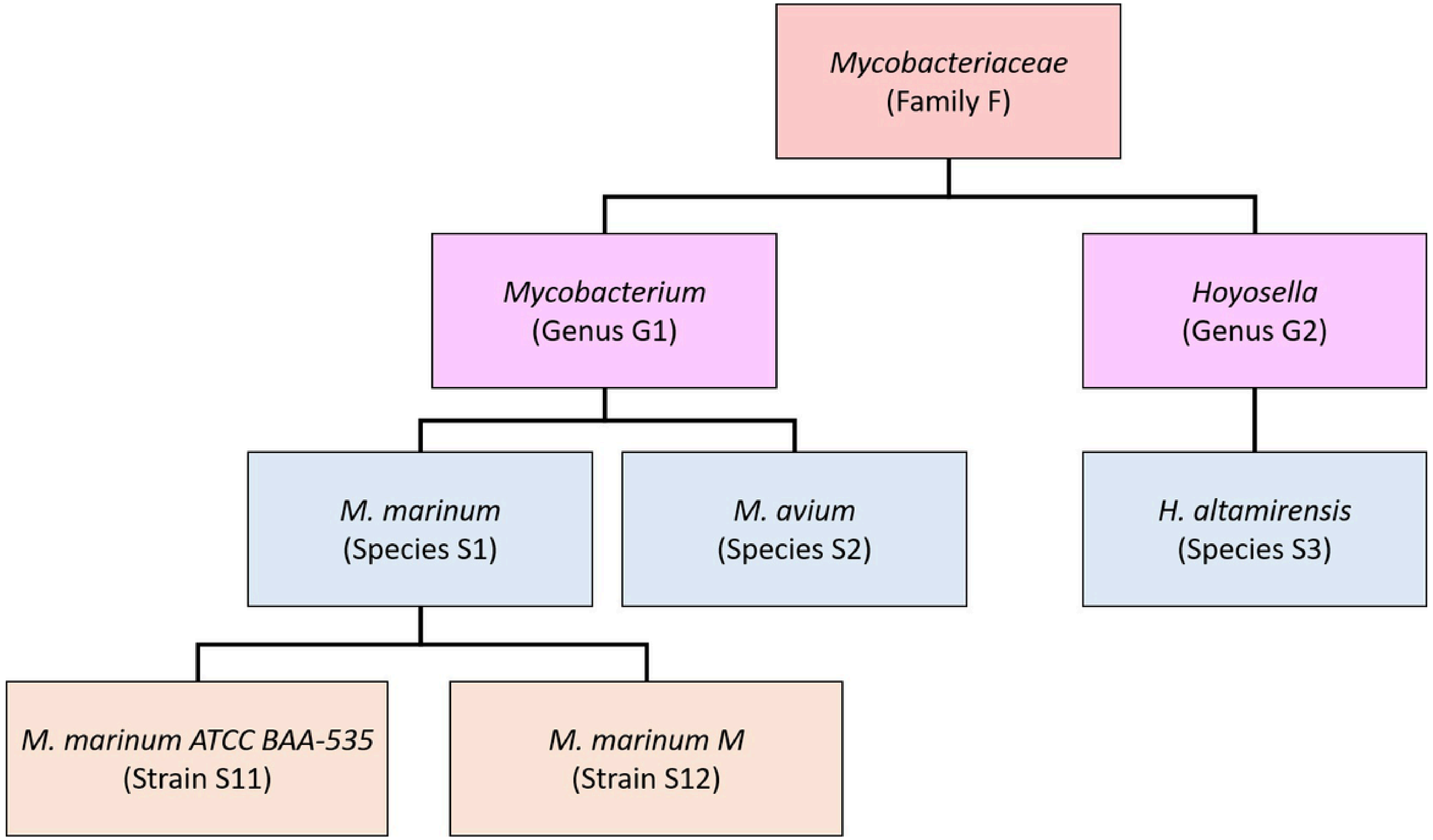


Output

```
78.13  587119  587119  U       0       unclassified
21.87  164308  1166    R       1       root
21.64  162584  0       R1      131567    cellular organisms
21.64  162584  3225    D       2           Bacteria
18.21  136871  3411    P       1224          Proteobacteria
14.21  106746  3663    C       28211           Alphaproteobacteria
 7.71  57950   21      O       204455            Rhodobacterales
 7.66  57527   6551    F       31989               Rhodobacteraceae
 1.23  9235    420     G       1060                  Rhodobacter
 0.76  5733    4446    S       1063                    Rhodobacter sphaeroides
```

1. **Percentage of reads covered by the clade rooted at this taxon.**
2. **Number of reads covered by the clade rooted at this taxon.**
3. **Number of reads assigned directly to this taxon.**
4. **A taxonomy rank code.**
5. **NCBI taxonomy ID.**
6. **Indented scientific name.**

Bracken: Bayesian Reestimation of Abundance with KrakEN



Mycobacteriaceae
(Family F)

Mycobacterium
(Genus G1)

Hoyosella
(Genus G2)

M. marinum
(Species S1)

M. avium
(Species S2)

H. altamirensis
(Species S3)

M. marinum ATCC BAA-535
(Strain S11)

M. marinum M
(Strain S12)

$$P\left(S_i|G_j\right) = \frac{P\left(G_j|S_i\right)P\left(S_i\right)}{P\left(G_j\right)}.$$

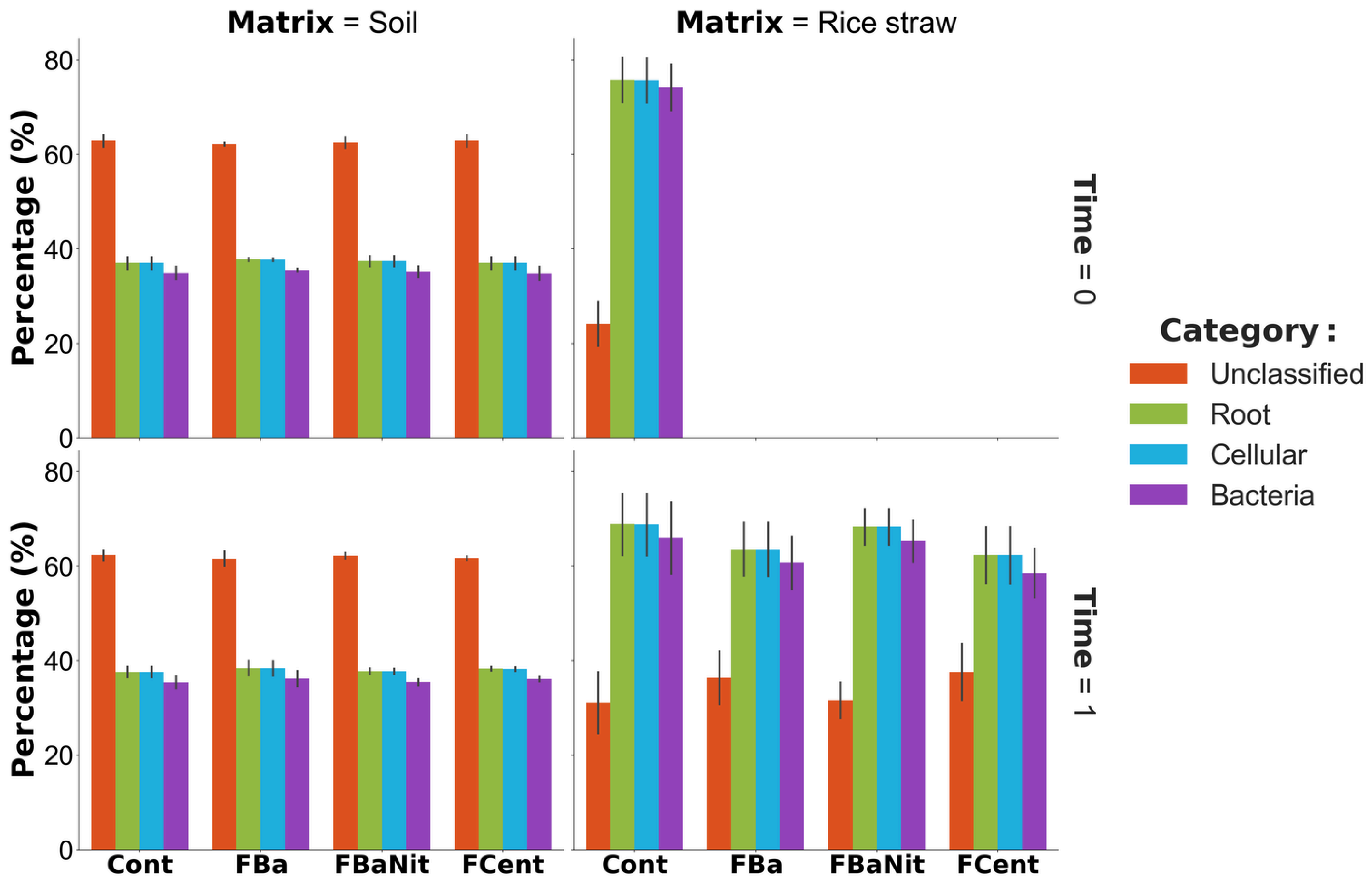
$P(S_i|G_j)$: probability that a read classified at genus *Gj* belongs to the genome *Si*. (**Posterior**)

$P(G_j|S_i)$: probability that a read from genome *Si* is classified by Kraken as the parent genus *Gj*. (**Likelihood**)

$P(S_i)$: probability that a read in the sample belongs to genome *Si*. (**Prior**)

$P(G_i)$: probability that a read is classified by Kraken at the genus level *Gj*. (**Marginalization**)

Example:



Problems with databases:

- Gap with the real world diversity.
- Overrepresentation of species.
- Low-abundance species.
- Database corruption.

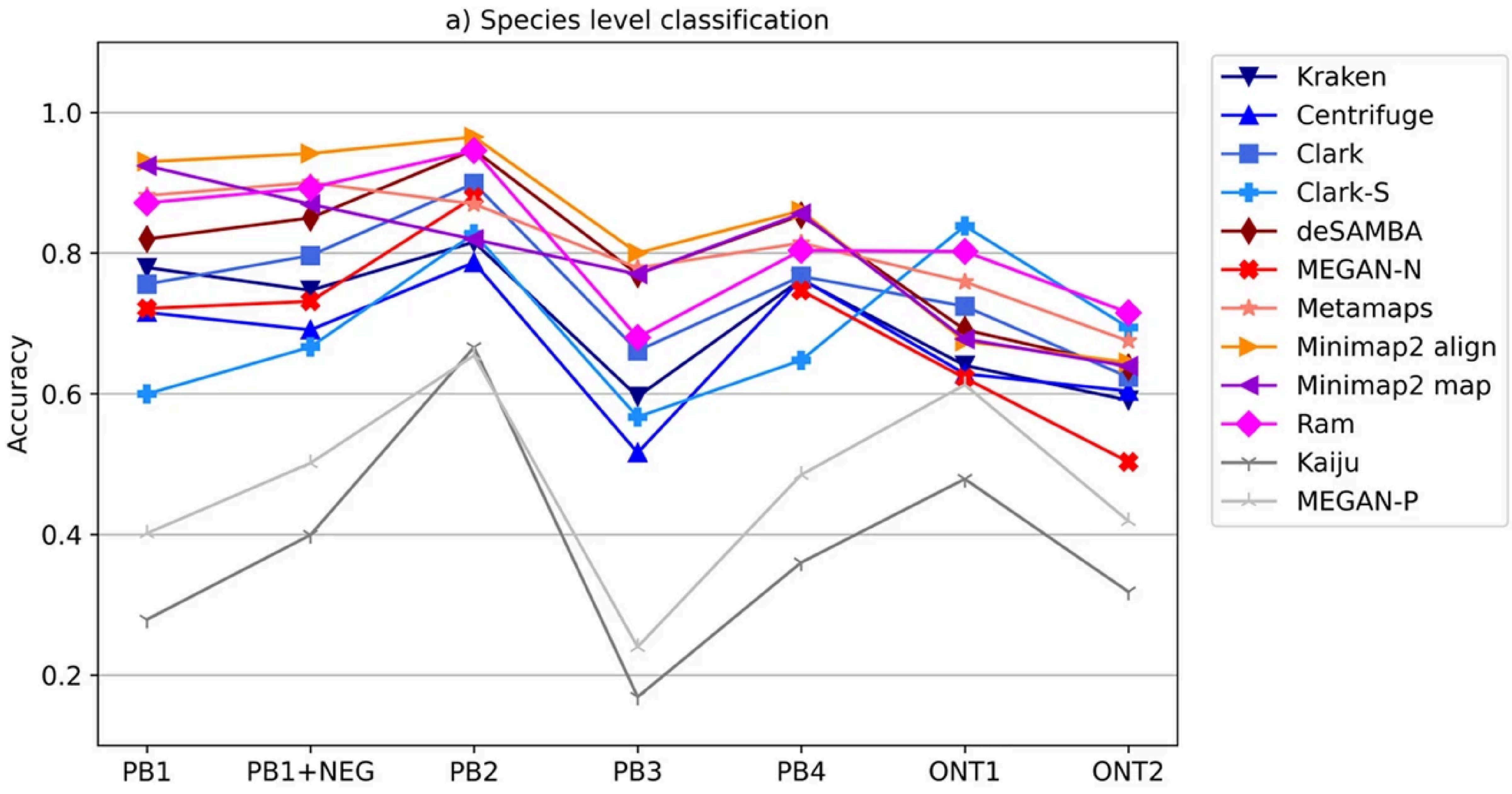
Short reads **vs long reads:**

- More informative per read → thousands of k-mers or long alignment regions.
- Improved taxonomic resolution (often genus/species level).
- Better handling of repetitive regions and structural variation.

However,

- Sequencing errors disrupt k-mers, reducing match accuracy.
- Lower classification rates if many k-mers are corrupted.
- Over-assign fine-rank taxonomic labels (sparse or localized k-mer matches).

Long-read classification:



Marić et al. (2024)

Specific tools:

Lemur/Magnet

Melon

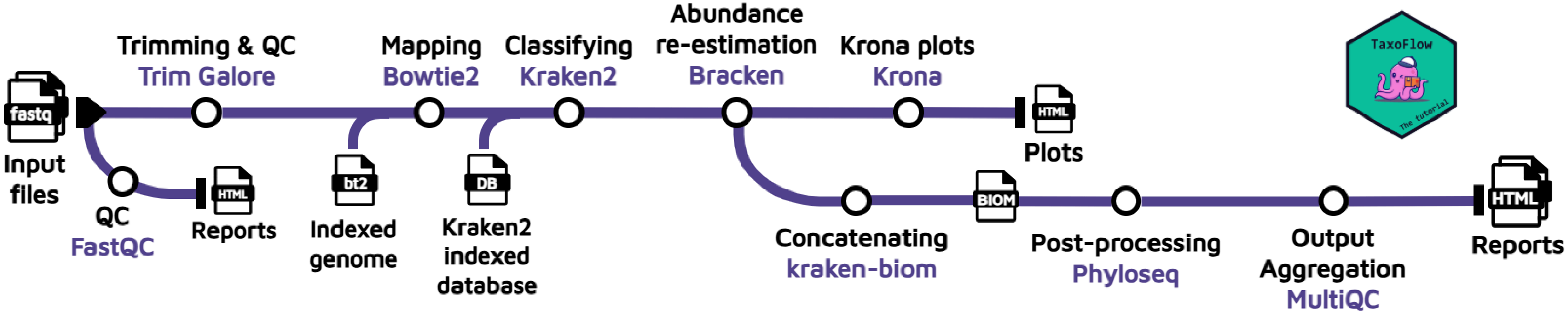
Today’s data:

The dataset we will be using for this part of the workshop was collected in **Cuatro Ciénegas (México)**. Cuatro Ciénegas is an oasis in the Mexican desert whose environmental conditions are often linked to the ones present in ancient seas, due to a **higher-than-average content of sulfur and magnesium** but a **lower concentrations of phosphorus and other nutrients** (Okie et al. 2020); the BioProject accession number: PRJEB22811.

Name	Group	Replicate
ERR2143758	Cont	1
ERR2143759	Cont	2
ERR2143760	Cont	3
ERR2143771	Unenriched	1
ERR2143772	Unenriched	2
ERR2143774	Unenriched	3
ERR2143769	Fertilized	1
ERR2143770	Fertilized	2
ERR2143773	Fertilized	3

TaxoFlow:

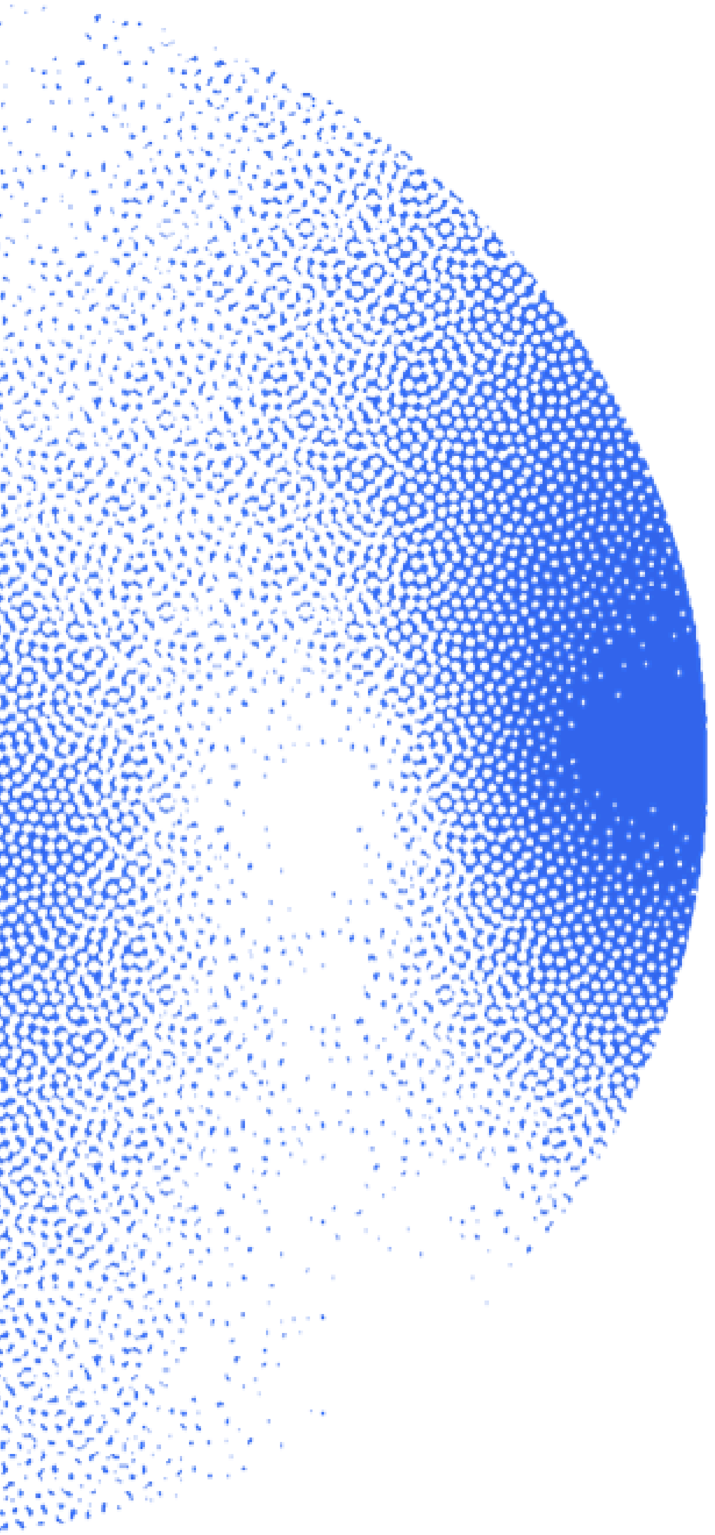
For this course, we propose to wrap with Nextflow the protocol published by Lu et al. (2022) wrapped with Nextflow:

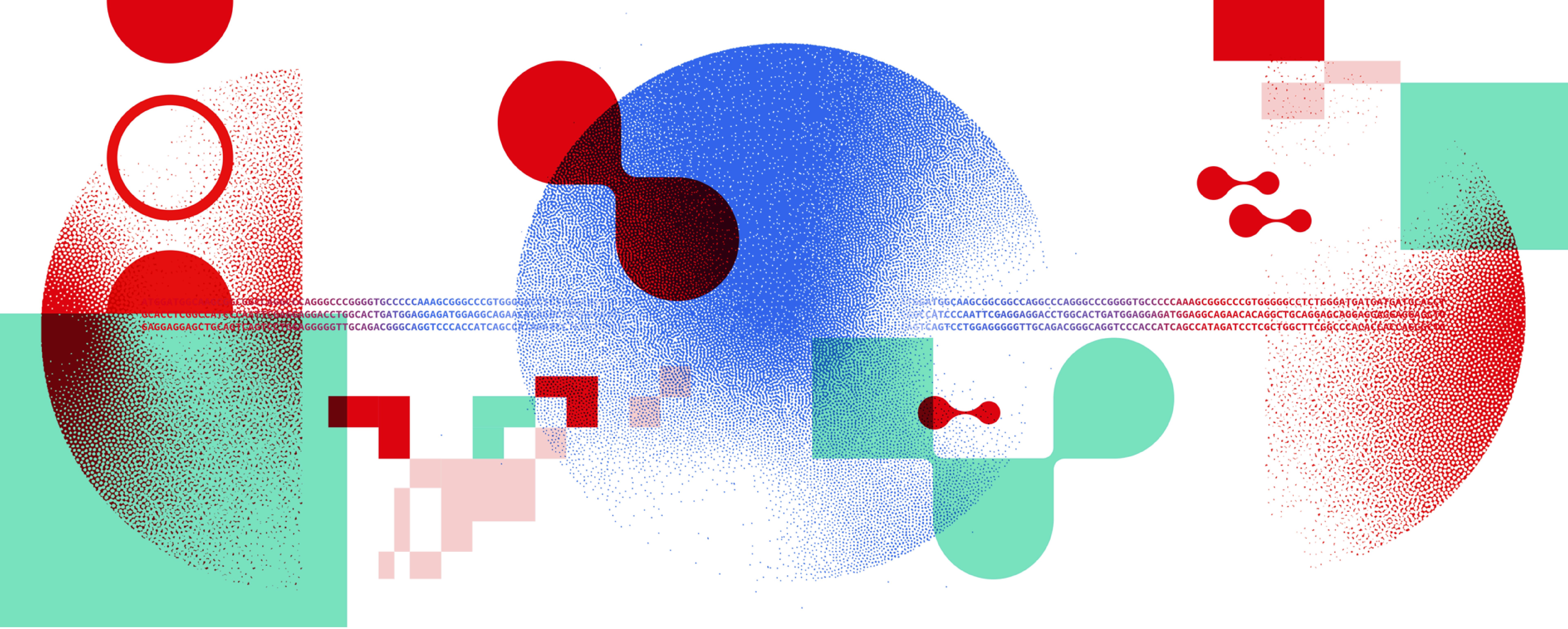

fastq
Input files
Trimming & QC
Trim Galore
QC
FastQC
HTML
Reports
bt2
Indexed genome
Mapping
Bowtie2
DB
Kraken2 indexed database
Classifying
Kraken2
Abundance re-estimation
Bracken
Krona plots
Krona
HTML
Plots
Concatenating
kraken-biom
BIOM
Post-processing
Phyloseq
TaxoFlow
The tutorial
Output Aggregation
MultiQC
HTML
Reports

Summary

- Contaminant removal is a key step during read-based metagenomics data analysis.
- Several options to analyze unassembled reads from a taxonomic perspective.
- Current state: “your analysis is as comprehensive as your database”.
- Long reads can capture more information, although they are more suitable for assembly-based analysis.

Practicals





Thank you